Jinhong Yu

Master's degree in Artificial Intelligence

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Summary

Robotics Software Engineer with experience in robot teleoperation, motion control, and XR-based human–robot interaction. Skilled in developing end-to-end robotics data collection systems using VR headsets and handheld devices. Experienced in fine-tuning and deploying AI models, particularly robot policy models, as well as diffusion and NLP models, with additional experience in web development.

Work History

Full-Stack Software Engineer , SwiftShop (Aug. 2026 - Present)

- Develop an e-commerce platform with a store management backend and a customer-facing WeChat Mini Program.

Founding Robotics Engineer, Theta-intelligence (Sep. 2024 - Jul. 2026)

- Developed a full-stack VR teleoperation platform for robotic systems, including interactive user interfaces, mixed reality (MR), computer vision integration, and external UVC camera support.
- Developed Android native plugins for Meta Quest devices to integrate ONNX-based AI models, including on-device text-to-speech (sherpa-onnx) and other edge AI applications.
- Designed and integrated hardware systems for robot data collection, including Meta Quest-based UMI and egocentric sensing platforms.
- Built end-to-end data collection pipelines using Raspberry Pi and Meta Quest devices, enabling synchronized recording of video, sensor, and IoT data for robot learning applications.
- Developed real-time communication infrastructure for multi-modal media streaming between robots, XR devices, and remote operator stations.
- Developed AI-enabled embedded systems on ESP32 (ESP-Claw) and Raspberry Pi (zero-Claw) platforms, deploying a voice-interactive AI agent for natural language communication and robot control.

AI Model Engineering Intern, Cardinal Operations (May. 2024 - Oct. 2024)

- Provide coding support to the text-to-image/video experimental project.
- Finetune Stable Diffusion model with limited hardware resources with LoRA and model parallelism.
- Deploy text-to-video pipeline, focusing on multi-character generation and character consistency

Research and Teaching Assistant, Northeastern University (Jun. 2023 - Dec. 2023)

- Homeworks, office hours. Conducted a guest lecture in diffusion model and gained a good feedback.
- Deployed servers in the lab to accelerate research projects. Participated in various research initiatives by designing, implementing, and running experiments, enabling project leaders to concentrate on idea development.

Skills & Knowledge

- **Robotic:** ros2, dora
- **XR development:** Unity, Open XR framework
- **Hardware:** 3D modeling, 3D printing
- **Embedded Development :** ESP-IDF, sensors and motors
- **Web :** Spring boot, vue, SQL server, Redis, Caddy
- **AI coding:** cursor, claude, with mattpocock skills

Education

Master of Science in Artificial Intelligence,

- *Northeastern University, Khoury College of Computer Sciences, 2021-2023*

Bachelor of Engineering in Machine Design & Manufacturing and Their Automation

- *South China Agricultural University, College of Engineering, 2013-2017*